

Class Project U4.

Cart-Pole Reinforcement Learning Project

In this project, the students will explore the Cart-Pole Reinforcement Learning problem and develop a complete technical guide describing how to configure and execute the environment step by step from scratch.

As part of the investigation process, students are encouraged to read, study and reference the first four chapters of the book *Deep Reinforcement Learning Hands-On* by Maxim Lapan. As well as any additional resources you consider useful.

The main objective of this assignment is to create a technically well-structured document that explains, step-by-step, the complete setup process required to prepare the development environment and run the Cart-Pole problem.

Deliverables

1. A PDF document containing the complete technical guide and implementation process. The document may generally include the following topics:
 - Basic explanation of the Reinforcement Learning workflow used in the project.
 - Description of the Cart-Pole environment and its objective.
 - Code organization and explanations.
 - Screenshots, execution results or plots.
2. A folder that contains the Python scripts properly named to run sequentially.

Deadline

The final project must be submitted by email no later than May 28th, 2026.

Final suggestions

Students should present the project as a reproducible technical implementation guide written from their own understanding and perspective. The document must demonstrate clarity, technical organization, and detail to reproduce the Cart-Pole implementation successfully.